**Academic research: Correlation Research between a User’s age and their Personally Created Avatar’s Characteristics.**

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**Abstract**: With an increasing amount of time spent online whereby the users represent themselves to their peers via custom avatars, it is crucial to outline how the user's perception of avatars relates to the avatars they create. This research is trying to find a correlation between user age and their personally chosen avatar characteristics. The act of designing a custom avatar to present oneself is by itself a reflection of what the user expects to encounter as a result of the choices the user makes. This anticipation is hypothesized to be related to the extent of experience a user has with the creation process and use of these avatars in different settings, which in turn is expected to be correlated with their age. A found correlation between user age and avatar characteristics has potential for prediction modeling and monetization purposes for gaming platform service providers. This research leverages repertory grids to gauge the emotion drivers of users to identify any correlation trend between a user's age and their avatar characteristics from interviews and ratings. The findings have found no linear correlation between user age and personally chosen avatar characteristics. This result was due to data-size and not noise, a larger and deeper dataset is required to shed more light on said relationship. Thus, this paper concludes that age is not enough to profile user perception of avatars. However, repertory grids have been found as a good basis for quantifying the emotional drivers of users from the low rate of noise created.

*Keywords*: Correlation, avatar, avatar characteristic, age, coefficient, personal

# 1. Introduction

Video games, consoles, massively multiplayer online role-playing games (MMORPGs) and gaming platforms of any kind are built and run by service providers. A service provider is an organization that provides IT solutions/ services to users and organizations (Barth et al., 2011; Gaming Control Board, 2019). Service providers can be used as a blanket term for a lot of companies that deliver many variations of services. The service providers this study refers to are those offering access to centrally hosted multiplayer games and are compatible with the Ready Player Me (RPM) tools to create custom avatars. Examples include apps like VR chat, Nemesis, Wonda VR, Sidequest.

Service providers hosting multiplayer games have in general two types of revenue streams. Some request a subscription fee and others offer options to buy upgrades and items in their platforms or a combination of the two (Fourberg et al., 2021; Kox, Straathof & Zwart, 2016; Ullah, Boreli & Kanhere, 2020). The latter is a strong revenue stream as it does not increase the entry barrier for the user to start using the service and it leverages a good avenue to trigger impulse buys (Clement, 2022a; Statista, 2022). As a result, 'in-app' purchases will be very important to the business model of service providers of gaming platforms. To increase in-app purchases they rely heavily on targeted advertising models to provide their users with appealing offers to upsell and engage more on their platform (Lashbrook, 2021). Examples relate to loot boxes, personalized avatars, additional currency or an ad-free experience.

Targeted advertising models rely on the knowledge collected from users and are always seeking new innovative avenues to find out more about the user and his/her interests. The more time users spend on the platform, service providers acquire more opportunities to get to know the user better. This leads service providers to come up with plans to increase user in-app purchases. For instance, studies on user behavior have shown higher levels of engagement in relation to making their own personalized avatars as they identify more strongly with it (Hemp, 2014). The ability to personalize avatars is key in enhancing a user's retention to the platform game and advertising (Hota & Derbaix, 2014; Kim & Sundar, 2020; Wu, 2014).

Since avatars can be used as vehicles for self-expression, entertainment, and escapism, service providers have the chance to profile users based on their choice of avatar characteristics (Hess, 2016; Rahill & Sebrechts, 2021). Creating virtual identities is not a universally shared experience. People who grew up making virtual identities will have a different perspective than people who have not. As a result, avatars are perceived and experienced differently by users, which in turn affects the choices of avatar characteristics (Kelly, 2017).

Personal construct psychology is a theory that stipulates a person's action is determined by their anticipation/expectation of a certain outcome which is, in turn, based on their perception of the world/topic and can be understood from the bi-polar constructs they use (Walker & Winter, 2007). Therefore, based on this theory, users customize avatars based on how they anticipate the choices they make will affect their experience using the avatars in a particular platform context. These users' choices when creating an avatar offer the opportunity to profile users. Since a person's anticipation of the consequences of their choices is directly linked to their level of exposure/ experience with how these choices will have an effect, personal construct theory suggests that there is a correlation with age.

Which is why this academic research wants to study to see if there is a correlation between a users' age and their chosen avatar characteristics in gaming platforms. Please note that avatar characteristics in this paper refers to the differentiating aspects of an avatar in the way that users perceive it. During the process of creating their avatar, users will make choices in relation to hairstyle, clothes, facial features leading to a final version of their avatar whose certain characteristics will be perceived in a certain way. Examples of these perceived avatar characteristics are heroic, villainous, confident, shy. These characteristics are of interest in this study as they are the differentiating elements in how the avatar is perceived by its creator (driven by the anticipation of the consequences of its creator).

It should be noted that this correlation research project is meant to be a proof of concept to not only service providers but also other analysts to further research.

## 1.1. Problem definition

Target advertising is based on big data analytics, where user's personal information is collected and processed for the purposes of profiling and targeting (Ullah, Boreli & Kanhere, 2020). Websites and services providers like Google and Facebook have created targeted advertising through the compilation of user's data, i.e. their likes, interests, history, geo-location and purchasing behavior to determine what specific ads are best suited for this specific user

(Cullfiord, 2021; Ives, 2021; Rotter, 2018). Targeted advertising relies on a platform's community of users spending a lot of time on the said platform as a result of high engagement (Alkan & Daly, 2020). High engagement is tied with the gaming service offered, in conjunction with a successful targeted advertising. Hence, it's a balancing act between the two. If the targeted offerings do not meet the interests of the user, the user will be annoyed, limiting the engagement (Wu, 2014). It is therefore very important to have a high quality user-profile to drive an effective advertising model (Blichev & Marston, 2003; Kox, Straathof & Zwart, 2016). In the context of gaming platforms, a good user experience is paramount.

Studies on the social aspects of platforms such as user ambitions, desires and needs are the most difficult to capture and there is always a need for better prediction models over time (Alkan & Daly, 2020; Eke et al., 2019). However, these exact aspects are expected to have a high value when the user places him/herself in a new world. The user-created avatar could be an essential indicator of these aspects of the user-profile.

According to the theory of personal construct psychology, people make choices based on the anticipated effects they expect to encounter, framed by experiences developed over time (Kelly, 2017). It follows that this would also be the case in relation to the choices they make when designing their avatar. A user designing an avatar is framed by the extent of their past experience with the process of creating/using avatars in particular situations (such as different game settings, platform context, avatar characteristic options). All of which is, in turn, is expected to be correlated with their age. Video games, and avatar customization, are more popular and available to present-day generations than previous ones (Clement, 2021; Statista, 2022; Forsans, 2020; Kristanto, 2018; Wallach, 2020). Hence, the creation of avatars between the age groups will be different by virtue of different experiences with the creation and use of avatars. A user that has created hundreds of avatars will create their next avatar more differently than one that has never created one before.

Finding a correlation between users age and avatar characteristics requires determining the specific characteristics of the avatar to use in this research. With a reasonable set of avatar characteristics, the correlation needs to be proven.

## 1.2. Research Objective/ Central Research question

Research objective: To determine whether there is a possible correlation between a user's age and personally chosen avatars characteristics of an avatar that user created under a specific context.

**Central Research question:** Do the properties of an avatar created by users provide for an efficient way to determine the users age?

**Sub questions:** To answer the central question, this report will be focusing on these two areas: service providers and user profiles. All these questions are necessary to first contextualize and clarify what past research has found in relation to correlation research in avatars and user profiling. Area 1 is about contextualizing past literature on correlation and prediction research in the gaming industry. Area 2 is about researching what influences a users' choices on avatar characteristics. This is all meant to find a quantifiable method that gauges the emotion-drivers of users towards avatars.

Area 1: The correlation research design on users (Service providers, user profiling, predictive modeling)

1. What is targeting advertising and user profiling?
2. How is user profiling built?
3. What are the current uses of user profiling and targeted advertising in the gaming industry?
4. What is the importance of predictive modeling?

Area 2: The relationship between user age and chosen avatar characteristics (User age, profiles, user motives, avatars and avatar customization)

5. What are the motives of users in creating an avatar?
6. How does platform context affect user choice of avatar characteristics?
7. How can a user's choice be quantified despite platform context?
8. How does user age affect choices of avatar customization?

## 1.3. Relevance

If the research demonstrates a way to predict the users age based on the characteristics of the avatar created by the user, then this opens an avenue to more extensive research on what other user profile properties can be determined from the avatar characteristics in support of a more advanced user profile assessment/ target advertising. Service providers can also better find and segment specific age groups through choice of avatar characteristics and formulate plans of action to engage for each age demographic.

Academically, this could shed more light on the research on the relationship between users and avatars and what this possible correlated relationship could mean to users. For the industry, such a correlation provides new avenues for not just service providers but all kinds of developers to better target users with relevant (buying) options on their own gaming platforms.

### 1.4. Research Line

The research line has started with what was the relationship between users and why they choose certain avatars in different platforms. Previous research has indicated that it all depends on the context of the platform (is it a gaming chat in a fantasy setting? Or is it a social-interactive setting to connect with friends and family?) (Hepperle, 2021; Kao, Harrel, 2018). Having the ability to create your own personalized avatars in almost every type of platform has led to a higher level engagement with the platform (Fox & Ahn, 2012; Kao, Harrel, 2018; Rahill & Sebrechts. 2021). The ability to personalize and choose your own kind of avatar gives the user a sense of “ownership” and control over who they can be on the platform (Turkay, Kinzer, 2016; Waltemate et al., 2018). This gives users a sense of agency through their avatar which results in users being more engaged (Dechant et al., 2021, Kim & Sundar, 2012). An increased engagement translates into an increased amount of time spent on the platform and leads to a higher success rate of ‘in-app’ purchases.

Avatars hold a special place in the minds of users on platforms but their choices they made to design their avatars reveal aspects about themselves and are not yet quantified (Antoniou et al., 2013; Kim & Sundar, 2012; Koles & Nagy, 2012). Therefore if there is a potential correlation between avatar characteristics (hair, height, clothes, eye color) and their real-life user profile, then developers and service providers have a new opportunity to better target their users. User profile aspects that may come into play are gender, sexual orientation, age, income, weight, but also agreeableness, conscientiousness, openness, neuroticism, and extraversion (Fong & Mar, 2015; SlideModel, 2020).

To properly test the correlation between the characteristics of a user-created avatar and the user, tools are needed where participants can easily generate avatars in order to collect the properties of both the avatar and the user.

Ready Player Me offers this platform and is compatible with many different types of gaming platforms, which also allows for future research to compare and contrast the effects of avatars in different platform contexts. See link for avatar creation tool in appendix 1.

# 2. Literature Review

## 2.1. Service providers

### Targeted advertising

Target advertising is about ensuring increased clicks and user engagement on a platform through extracting, profiling and categorizing a user's personal information to personalized ads (Ullah, Boreli & Kanhere, 2020). The entire process of targeted advertising starts with using user profiling to categorize user profiles with personalized ads and deliver said ads to the user's device based on time-sensitive moments in the day (Alkan & Daly, 2020; Eke et al., 2019; Spychka, 2022; Ullah, Boreli & Kanhere, 2020).

Ads are everywhere on the internet. Americans see on average 3000 advertisements a day (Anand & Shachar, 2009). But no one can remember every ad, which is detrimental to advertisers as the most important aspect in an online environment is being seen. This is the exact problem that targeted advertising is meant to solve by profiling what the user wants with personalized ads. The greatest effect of advertising is done by targeted advertising via segmenting specific ads to interested users (Melnyk & Kirkin, 2017). That is the importance of target advertising and the results have been very profitable: Targeted advertising most notably helps account for the total gaming industry number of 197 billion dollars through the mobile market (Clement, 2021; Clement, 2022c; Statista, 2022). Targeted advertising and in-app purchasing is most popular in free-to-play platforms with revenues totaling 106 billion dollars (Clement, 2022b, Clement, 2022d). In-game spending options are a very viable source of revenue for gaming service providers.

A more accurate user profile is liable to create a more effective target advertising model which, in theory, leads to more engaged users as well as potential advertisers. The more a user interacts with a certain source or platform, the data that can be extracted for profiling (Eke et al., 2019). Hence, creating a cyclical system for a good business model for service providers as well as an incentive to extract as much data as possible from users for an effective targeted advertising system.

It should be noted that induced purchasing intent from targeted advertising in gaming platforms function differently compared to social media platforms. Social media engages users with more social-based motivators: friends, family, personal advertisements to real life problems, convenience and passing the time (Alhabash & Ma, 2017; Bowden-Green, Hinds & Joinson, 2021). Gamers are engaged to purchase based on game-related motives that are not typically related to their real-life problems (Hota & Derbaix, 2016; Jimenez et al., 2019). The success factors for service providers on gaming platforms is the level of user engagement/ retention and monetization (Hadiji, et al, 2014).

A key for a successful business model on a gaming platform starts with having the trust from the users. If there is no trust from the users, they are unlikely to engage with the platform, especially with personalized content (Chellappa & Sin, 2005). Trust determines how much interaction and open a user is to allow a platform to data mine for accurate profiling. Under this context, service providers must trend the fine line between monetizing their content and maintaining their user trust to ensure higher levels of engaged users over time. The more users are engaged, the better modeling for targeting advertising is made.

## User profiling

User profiling is referred to in this study as collecting data on a user's interests, characteristics, behaviors and preferences (Eke et al., 2019). Data is accumulated from a variety of different sources and channels to create profiles. User profiling was developed to condense large amounts of user data to use for marketing and product design. Analytics, service providers and marketers can use said user profiling to draw inferences between their platform properties and user profile characteristics (Gao, Lui & Wu, 2010). Such characteristics include behavior, age and income. These inferences between users characteristics/ data and platform properties is used to establish a plan of action to meet the needs of their users (Sifa, Drachen & Bauckhage, 2018).

The two ways to establish a relationship between platform properties and user profile characteristics is done by either correlational design or inferential design. Correlation design is seeing what can be inferred between user profile characteristics against platform properties that can range from user experience to monetization, user retention to cross-game transportation. Inferential design investigates how and why a specific behavior on a platform occurs as a result of the user's profile/ traits (Sifa, Drachen & Bauckhage, 2018).

The benefits that service providers could gain from user profiling is that it can act as a stepping stone to better know everything about their users for monetization and engagement purposes. Therefore, accurate user profiling is key for service providers. The more accurate, the better the chance of creating a successful business model. It is in their interest to know all of the characteristics of a user like age, income, socio-economic status, behaviors, habits and psychology (Canossa & Drachen, 2009; Fourberg et al., 2020).

Businesses of any and all kinds that plan based on scale and growth are heavily based on certainty (Dharouge, 2021; Hill, 2017). Data-driven research from profiling users creates conclusive certainty for better plans to either market or engage more users over time to the service provider's platform. That is how important user profiling can be to service providers and marketers.

## Building and current uses of user profiling

Building user profiling requires a goal, or need, in mind. These goals typically relate to collecting user data to discover and meet the needs of their users for engagement purposes. However, there are many approaches to build user profiling models in a gaming context. Examples of some of the approaches are called dynamic profiling, snapshot profiling, spatio-temporal profiling, predictive future behavior profiling, (Sifa, Drachen & Bauckhage, 2018). Dynamic profiling uses user data to find any change in behavior and needs over time (Drachen et al., 2016). Games change and update over time, it is best to know how this affects the user population. Snapshot profiling by contrast generates a user profile based on past historical data. This type of profiling is useful for establishing patterns of behavior to better categorize how users generally interact on a platform. Spatio-temporal profiling doesn't rely on historical data but how the user reacts in a virtual environment (Pirker et al., 2016). Not every game has the same environment or mechanics, it is crucial to see how users react to game environments to identify any potential pain points to engagement. Predictive profiling is about predicting future users behavior on the platform. Predicting future behavior helps identify what engages players and what fraction is willing to spend money on a platform (Sifa et al., 2015). There are many other different types of user profiling in a gaming context that overlap with each other (Sifa, Drachen & Bauckhage, 2018).

These types of user profiling methods can be used to answer questions such as “how engaged in the platform was the user in the past 3 months?” (dynamic) or “how has the newly added game arenas affected the different user demographics in terms of user experience?” (spatio-temporal). With no goal to meet a certain need in mind, a service provider is more liable to waste resources. User profiling is not a simple process, there are decisions and areas to be considered to best create conditions of a good user profiling model. For example, user profiling can be done either individually or by group. The grouping profiling is more common as user profiling means to condense large amounts of all users in a platform that can be used to gauge general sentiments and patterns of behavior. But there is a cost-benefit when compared to individual profiling as it is less precise when pin-pointing specific user characteristics.

Since user profiles are heavily data-based, service providers need user profiles that are data-driven (Sifa, Drachen & Bauckhage, 2018). Data-driven profiles are based on quantitative data. Data-driven profiles require constant updates in design and user data to remain current.

User profiles can and have been made through a wide range of methods/ inputs. For example, user profiles can be built by indirect and direct inputs (Gao, Lui & Wu, 2010). Indirect inputs extract user data, without the user’s awareness, from the web, personal devices, mobile networks or social networks (Cote, 2021). The direct method is where user data is extracted directly from the users themselves via surveys and questionnaires (Eke et al., 2019, pg. 5). All users have a vested interest in a ‘good’ user experience, since they, at the bare minimum, wish to be entertained. Although statistically, surveys typically nets low response rates ($n < 45\%$) of feedback, it is no less important in establishing user needs (Baruch & Holtom, 2008; Wu. Zhao & Fils-Aime, 2022). By providing their own user data and feedback to service providers, there is more potential in the creation of a better user experience. A better user experience means more engaged users, which potentially opens up the way for more effective personalization (Antoniou et al., 2013). Thus it’ll become a cyclical process.

A user's overall experience (needs, behaviors, psychology, and other user profile aspects) is most important to developing a plan to build a user profile (Sifa, Drachen & Bauckhage, 2018). The better a service provider meets the needs of the users, the more engaged the user will be (Alkan & Daly, 2020). The more engaged the user is, the more accurate a user profile that a service provider can make (Kim & Sullivan, 2019; Osten, 2022).

## Predictive modeling

Predictive modeling is about predicting user traits in the user for the future (Alkan & Daly, 2020). In behavioral user profiling, predictive modeling is used to predict specific behaviors in users at certain times during an experience on the platform (Sifa, Drachen & Bauckhage, 2018). Predictive modeling is most useful as they are key for business intelligence to make an actionable plan for service providers (Lawton, Carew & Burns, 2022).

An overview on one of the processes of predictive modeling is by first answering questions such as how many players will join this specific gaming platform by the 3rd quarter? The predictive modeling will take a large quantitative sampling of the platform's historical data on how many new players and those who remained. Once the model has found a pattern and made their predictions, their reliability can be judged over time with what was actually recorded until the 3rd quarter of this year (Chen & Kaufmann, 2021; Hadiji et al., 2014; Sifa et al., 2015; Waljee et al., 2019).

Other types of gaming profiling (Snapshot, Dynamic, Spatio-temporal) are more focused on game-related mechanics (environment, rules, other players). Predictive modeling takes all these factors a step further by predicting what is to be expected in the future. The predicting of future attributes and behaviors of players has the added benefit of cutting costs and revenues (Sifa et al., 2015). This is due to the fact that prediction modeling provides service providers certainty in the long term, through data-driven insights, to form strategies for marketing, platform design, development, and resource planning (Hadiji et al., 2014; Sifa, Drachen & Bauckhage, 2018).

Finding and predicting user attributes is the core for any kind of data-driven game development for either monetization, platform design and profiling future users' needs (Alkan & Daly, 2020; Hadiji et al., 2014; Wei et al., 2017). If the predictive modeling of this current research is found, it can potentially be used by service providers to find a specific segment of players/users that are most likely to engage with their platform.

## 2.2. User Profile

### Users motives through avatars and avatar customization

An avatar is a self-made virtual identity by a person to interact and behave in an online context (Hemp, 2014; Lin & Wang, 2014; Koles & Nagy, 2012). This research refers to avatars as game characters under the control of users. A user's engagement to a platform stems from how much attached they are with their avatar (Kao, Harrel, 2018). Thus, for service providers, avatars are a key towards engaging new users on their platform (Nadeem et al., 2020). User motivations have been categorized and listed in different variations (Chu et al., 2021; Fong & Mar, 2015; Suznjevic & Matijasevic, 2010). One such categorization is realistic, ideal, fantasy and role-player (Wu, 2014):

The realistic users create their avatars to match their physical appearance as closely as possible. The ideal users create avatars similar to their own real-life appearance but with a more attractive representation by manipulating factors such as age and weight. The roleplayer user creates their avatars to step into another persona, world or real-life person. The fantasy users differ from the role-player users by trying to create an entirely original avatar that has no bearing to any person/ character, real or fictional.

Both realistic and ideal users make no separation between their identity as an avatar and as a real-life person. Studies on both fantasy & role-player users show little to no overlap between the user's and their avatar's real personality since the user typically creates more than one avatar (Lin & Wang, 2014). Which does touch on the other motivations that influence users choice of avatars characteristics: relationships, manipulation, immersion, escapism and achievement (Yee, 2006). All of these motivations are summed up as users desiring to either interact with others, use others for personal gain, enjoy being in another world/ person, escape from real-life problems and stresses and/or become powerful under the context of the platform.

These motives affect the avatar creation because avatars are, for users, a mode for self-presentation, both consciously and unconsciously. An extroverted person will be motivated to create an avatar to look like themselves while neurotic ones focused more on shielding aspects of their true selves in their avatars (Fong & Mar, 2015). There is a strong correlation between users who create an idealized avatar of themselves and users who have high levels of attachment to their avatar (Ducheneaut et al., 2009). Users' motives to choose between avatar characteristics are based on what they self-perceive themselves might become, would like to become and what they are afraid of becoming (Lin & Wang, 2014). This suggests that users choose aspects in an avatar according to a set of their current perspective, behaviors and past experiences (Canossa & Drachen, 2009)

Avatar customization helps a user's self-presentation on avatars through their freedom to choose. Choosing certain avatar characteristics has a psychological effect of 'ownership' that they like to associate with themselves (Turkay, Kinzer, 2016; Waltemate et al., 2018). This amount of choice and control is appealing to both kinds of users who wish to be either themselves or someone else. A wide variety of avatar characteristic options will appeal and improve the user experience of users with different motivations, personalities and behaviors (Ducheneaut et al., 2009; Fong & Mar, 2015; Koles & Nagy, 2012; Kristanto, 2018; Wu, 2014; Yee et al., 2009).

It should be noted that studies of user motivation on avatars have revealed that, statistically, most users are more likely to create avatars that look close to themselves than someone entirely else (Fong & Mar, 2015; Rahill & Sebrechts, 2021; Wallace & Maryott, 2009; Yee, 2006).

Avatars are, in essence, a tool for exploration of identities, relationships and self-presentation through avatar characteristics (Fong & Mar, 2015; Lin & Wang, 2014; Yee et al, 2009). They reveal that the relationship between user and an avatar makes identity fluid, not just towards others but to themselves (Hemp, 2014). Meaning, that a user's sense of identity can be influenced through interacting as their avatar (Koles & Nagy, 2012; Lin & Wang, 2014). Avatars are key to a satisfactory and engaging user experience for users, which leads to more time spent on the platform (Trepte & Reinecke, 2010). The more time spent on the platform, the more likely the user will get attached to their avatar which leads to strong customer loyalty (Nadeem et al., 2020).

## The relationship between users and platform context

The relationship between a user and an avatar overlaps with a user's relationship with the platform context, the game world, that the avatar inhabits. The context of a platform directly influences how a user customizes an avatar and interacts with it (Bonny & Castaneda, 2022;

Fong & Mar, 2015; Mawhorter et al., 2018; Turkay, Kinzer, 2016). What is meant by the context of a platform is game genre, world, rules and purposes set by the platform which creates certain expectations and tones for users in their avatars. For example, if a platform has the context of a dating platform, users will be inclined to create avatars in ways that will best fit the context of such a platform, i.e. making themselves look attractive. Platforms with game-specific context such as world of warcraft will by contrast have users create avatars based on other characteristics such as looking more intelligent (Lin & Wang, 2014; Vasalou, Joinson, 2009). In a job interview context, a user would be inclined to create a more realistic avatar. A user's choices of avatar creation would change if the platform context had changed (Wu, 2014). The changing of context indicates how users use their control of self-presentation through their avatars to emphasize a specific quality about themselves to others (Fox & Ahn, 2012).

This emphasis on self-presentation extends not just to the choice of avatar characteristics (appearance) but also to how a user interacts with others. Tattooed avatars are seen as more risk-takers or sensation seekers than those without (Fong & Mar, 2015). Strong male avatars tend to behave more aggressively than female avatars, hence pushing a perception onto users that male avatars are less conscientious and less open to new experiences. Female avatars, by contrast, are seen as less emotionally stable than male avatars. Sex cues enable social categorization in the virtual world (Bruzek, 2015). Examples of these interactions between users create the expectation to conform their behavior to be like their chosen avatar's characteristics (Ratan et al., 2018). The cultural and gendered lines from the real world are not separate from the platform's virtual world, but more apparent in it (Kim & Sundar, 2012). The amount of user interaction with other users correlates heavily with avatar customization on a platform (Mawhorter et al., 2018). The community on a platform matters just as much in making a good user's experience to the platform context as the game world.

Context of a platform informs on expectations, appearance and behaviors of an avatar by users, but this it's not a fixed relationship. It's a constant negotiation where the platform's context challenges a user's perception on the appropriateness of their choice of avatar characteristics (Koles & Nagy, 2012; Vasalou, Joinson, 2009). However, if the context or world of a platform makes a user's personal identity feel inadequate through their avatar, then that leads to less engagement to said platform (Ducheneaut et al., 2009; Loewen, Burris & Nacke, 2021; Wu, 2014). The interaction of the platform's context that affects the choices of users make with respect to the design of their avatars can be further explained by a theory called personal construct psychology.

## Personal construct psychology

Personal construct psychology is a theory based on the notion that people's perspectives, and therefore their actions, stem from a collection of opposing aspects of their observations (Walker & Winter, 2007). Or in other words, a set of variables that people use/ construct to perceive themselves, the world and others in a certain way (Feixas et al., 2021). A simple example of this notion is: you can't understand what "hot" is without knowing what "cold" is. The constructs that people (individuals) create for themselves to make sense of the world is very subjective.

Example: Jimmy is asked about trustworthiness and his response is: "Well, I am very trusting of people, whereas my parents are always suspicious of their motives." Through the word of "trustworthiness", he distinguishes it with the opposite: "suspicious" as well as how he perceives himself, others and the world around him: assigning himself as trusting and his parents as the opposite. The key aspect is that Jimmy took what he knows (himself, his parents) as ways to perceive, or construct, a world that makes sense to himself. Thereby making a construct.

The total set of all constructs is something that allows people to perceive the world, themselves and others. Repertory grid research is essentially a technique that tries to capture the main constructs used by the individual (Walker & Winter, 2007). The words people use to describe objects, events, themselves or any aspect of the world around them is indicative of bi-polar constructs they use to make sense of the world in the context of their efforts to anticipate what will happen in the future if they take certain actions or see how events transpire. How a platform's context works with this theory is best summed in the example following:

Example: If Robert perceives himself as a lover, a manager, a father and a dancer, then he would change his avatar accordingly per platform. LinkedIn is a work-based context platform, so he is likely to make his own avatar to emphasize himself as a manager. Twitter and Facebook are social-based context platforms, so Robert would be inclined to create his avatar to emphasize his self-perception as a lover and a father. Tik-Tok's platform context would push him towards making a dancer avatar profile of himself.

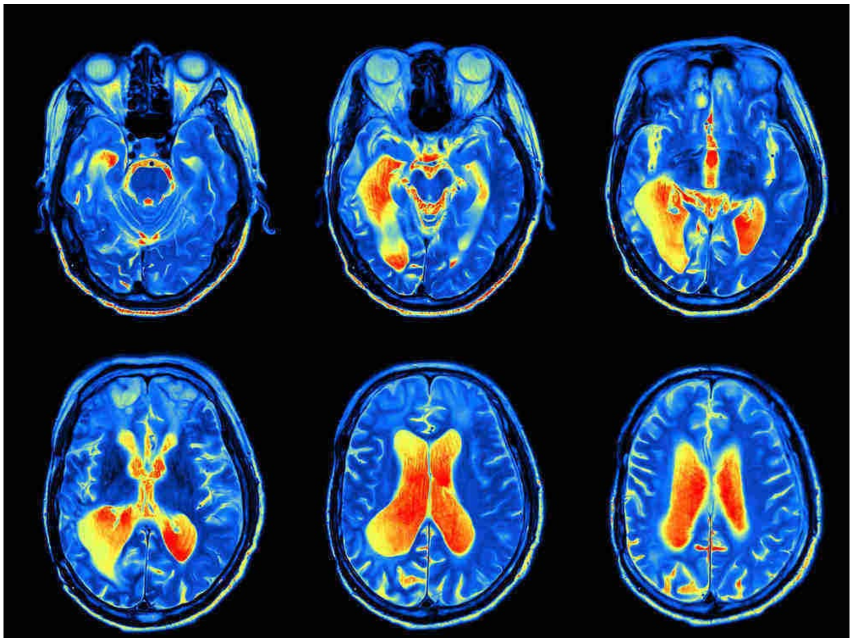
The importance of platform context is that it changes a user's choices, but it does not change their repertory grid (Fox & Ahn, 2012). Users use the platform's context to portray aspects of themselves that they deem most relevant to how they wish to be viewed on the platform. This is the similar case with avatar characteristics and gaming platforms. For instance, a game platform with a wild west context would present users choices between a criminal or lawman. Users would make the choice that they feel would best portray an aspect of themselves under the platform's context. These avatar choices will change if the gaming platform's context is science fiction.

Not everyone has the same overall repertory grids. Constructs are derived from recurring elements in a person's experience and are derived separately, thus making every person's overall perception of themselves, others and the world, unique (Carver & Scheier, 2000; Oosterwegel, 1995). Not every experience will be perceived/constructed the same by everyone and no experience can ever be fully recreated to the initial conception (Kelly, 2017).

Everyone constantly seeks to validate their own worldview/construct, so they'll surround themselves with people and things that do just that. This validation-seeking is found across all types of genders and ages (Walker & Winter, 2007; Valkenburg & Piotrowski, 2018). Repertory grids are not made in isolation, it is a learned construct from person to person. Just how a parent teaches their child to speak their first words, they help to identify who they are to each other: parent and child. A good example is in Figure 1 below:

**Figure 1**

*Image of a brain scan*



- This image was taken from the source: (Rege, 2022)

Figure 1 is a brain scan. Any person can just simply see the brain scan, but a trained neurologist can see this brain scan differently and diagnose that the patient has ADHD. A neurologist sees the same picture but has learnt how to differentiate particular aspects in relation to the diagnosis he/she wants to make. This is the result of a more refined repertory grid being taught by other specialists. A person's personal construct is just as much a self-made creation as it is the result of other people's confirmation (Walker & Winter, 2007). There is a shared learning curve between experienced and inexperienced constructs when centered on a specific topic (Bradshaw, et al, 1993; Roberts, 1999). The repertory grid of inexperienced people on a specific topic is a subset of people who have had a lot more experience with the topic at hand.

Repertory grids are determined by listing all of the attributes that were verbalized by participants to differentiate between objects (Goffin, Lemke & Koners, 2010). By interviewing a large group of people with diverse backgrounds, it is possible to collect a large spectrum of different attributes used to differentiate on a particular topic or object. By analyzing the unique and common responses, a designed set of attributes representing the most important ones can be used for quantitative study. Past studies have used the repertory grid technique for other purposes such as marketing and finding usability problems in customer's online experience (Heckmann & Burk, 2017). This technique can also be applied in relation to how people perceive avatars which influences the choices they will make when designing an avatar. This is due to repertory grids being a result of the past experiences and behaviors, which evolve over time. This is the case because of the amount of experience a user has had on a specific topic. Therefore, as a result of age, individuals build different repertory grids because of their experiences.

## The importance of age to users

Age is a good dividing factor between people, especially when gauging the differences between the generations (Gramlich, 2021; Pew Research Center, 2020; Pew Research Center, 2022; Schaeffer, 2021). Age stays a consistent user profile factor and is not as volatile as other factors such as user behavior (Sifa, Drachen & Bauckhage, 2018). Since everyone requires validation for their construct, age shows how different generations use trends, both past and present, to self-identify/ distinguish themselves from others through body language, slang, fashion, hygiene, nutrition and habits (Alkan & Daly, 2020).

Age, just like all avatar characteristics, are factors that can be changed in avatar customization by the user for the sake of user-representation. One study has shown that the age of an avatar matters more to older users compared to younger users (Ducheneaut et al., 2009). People, who are older, wish to be younger, whereas younger people wish to be older (Chopik et al., 2018). The motivations to changing age as an avatar characteristic are not always conscious to users (Ducheneaut et al., 2009). Since video games, and thus avatars, are eventually going to be used across all age groups in the long term due their current mainstream appeal, specific targeting strategies to different age groups are expected to form (Clement, 2021; Knezovic, 2022; Suznjevic & Matijasevic, 2010; Tolić, Šimunec & Vuković, 2020; Wallach, 2020; Yee, 2006).

Everybody views age differently as opposed to the actual age that they feel that they are. People who are aged between 18-64 think that a person is considered "old age" at 71, whereas people who are 65 or older think the age 77 is considered "old age". Males think "old age" starts at 72 whereas females think "old age" starts at 75 (Daignault, Wassef & Nguyen, 2021). The answer to when "old age" starts is arbitrary and varied between different demographics like race, gender, income and education. But this is indicative of a person's construct changing over time with age (Walker & Winter, 2007).

Repertory grids are defined by past experiences and behaviors. Age plays a big role as unique repertory grids from people are made over time per generation (Carver & Scheier, 2000; Oosterwegel, 1995). Younger generations are more exposed to avatar customization than older generations (Ofcom, 2021). Exposure is key to the importance of age as younger aged people are more impressionable than older age groups. Youths, ranging from adolescence to young adulthood, are in an early stage of development where they are susceptible to absorb almost anything to create their own sense of identity (Gwon & Jeong, 2018; Large et al., 2019; Nesi, Telzer, Prinstein, 2020). In the case of avatars, these young age groups who are exposed to creating avatars are more susceptible to self-identification with their own avatar than ones who are not used to it (Villani et al., 2016). This exposed age group will create a sense of identity as well as gain self-esteem and potentially other therapeutic effects through their personalized avatars (Dechant et al., 2021; Fong & Mar, 2015; Kao & Harrel; Kim & Sundarr, 2012). These effects from personalized avatars will affect the perception of avatars. This perception of avatars will translate into actions and that will be visible in avatars from both exposed and non-exposed age groups. Consequently, the choices to customize and interact as avatars are going to be different between younger and older age groups (Canossa & Drachen, 2009; Kelly, 2017; Weissman, 2017).

## 2.3. Hypotheses

It is hypothesized that there is a correlation strong enough to determine a user's age based on the chosen characteristics of their personally created avatar. Customized avatars are an extension of identity to their users on a platform (Kao & Harrel, 2018; Fox & Ahn, 2012). User profiles are a summary of a user's interests, characteristics, behaviors and preferences (Eke et al., 2019). Avatars are created based on past experiences, behaviors and perception of avatars. All of which are segmented by age through a difference in exposure to avatar customization among the generations (Kelly, 2017; Weissman, 2017). Younger generations have more exposure to avatar customization than older generations and therefore have a different perception of avatars (Knezovic, 2022; Ofcom, 2021; Suznjevic & Matijasevic, 2010; Wallach, 2020). Perception, behaviors and experience drives the actions (Canossa & Drachen, 2009).

These actions will be reflected onto created avatars. This current research is about reverse engineering this process from created avatars back to age.

# 3. Methodology

## 3.1. Research design

The research approach is correlation design: an observational study on the relationship between two or more variables using statistical data. The methodology is thus meant to meet the main goal of finding a correlation between users age and avatar characteristics. Any correlation that is found in this study will be concluded as either strong, weak or a non-existent correlation (Leverage, 2021). The establishment of a correlation requires a large amount of profiled user data collected from as many users as possible for accuracy (Eke et al., 2019). Therefore the methodology is split up in between the qualitative and quantitative phases. The qualitative phase is a set of interviews to determine a superset repertory grid. The quantitative phases follow after the interviews with a survey to collect user age and their personally created avatars on Ready Player Me's avatar creation engine.  A second survey will be conducted to rate those created avatars based on the superset repertory grid. The results will be put into a comma separated file (CSV) file for data analysis to find a correlation and establish whether a prediction model is feasible.

The context and variables of this current research will not be manipulated for comparisons to ensure consistency in answers. Participants using Ready Player Me's avatar customization will only have to create a virtual video game avatar. A specific gaming context will be given to participants during the avatar creation as reference for what they should create their avatar for (friends, fantasy, escapism). This context given to participants will be creating an avatar in a MMORPG context where they will interact and have adventures with friends from real life.

## 3.2. Data collection method

### 3.2.1. Interviews, determining a superset repertory grid

The interviews are the qualitative part of data collection. In order to build a list of characteristics that are commonly seen in the repertory grid when seeing an avatar, people of different ages were interviewed to describe how, in their perspective, one avatar appeared different from another avatar. To elicit strong contrasts, the participants were asked to pick one avatar they liked and another they disliked from a list of pre-existing avatars (Goffin, Lemke & Koners, 2010; Walker & Winter, 2007). This list of pre-existing avatars can be found at appendix 14. The features that they use to describe why one was different from the other were collected to be compared across the different participants (Boyle, 2005). The similarities and differences in a participant's interpretation of avatar characteristics are highlighted in the transcriptions at appendix 16.

There is a large overlap on how different participants verbalize their perception of avatars (Oosterwegel, 1995; Walker & Winter, 2007). Participants could identify avatars as "realistic", "plastic", or "friendly". These words taken from a sample group of participants of different ages can help create an overall, general repertory grid for all ages. This super-set repertory grid of all the perceived differences and similarities between avatars from the participants helps quantify the impression a person will have on any particular avatar.

The reason for a superset repertory grid is based on the fact that no personal construct is made in isolation. Personal constructs depend on personal relationships between individuals. The more a person has in common with another, the higher chance that they develop deeper relationships (Walker & Winter, 2007). Therefore, interviewing participants to get a repertory grid is not just a window to their own personal constructs, but the many other people associated with them and how they too differentiate between avatar characteristics (Boyle, 2005; Caldwell & Coshall, 2002; Easterby-Smith, Thorpe & Holman, 1996; Jankowicz, 2001;  Marsden & Littler, 2000; Pidzamecky & Oostveen, 2021; Senior, 1996; Storr, 2004; Wright & Cheung, 2007).

The resulting superset repertory grid made from the interviews can be viewed in findings. This superset will be used in the survey to rate created avatars from a different group of participants from a scale of 1-6 between opposing characteristics. For instance, if a rater were to rate between happy and sad on a scale of 1-6: a rating of 1 would mean that the rater perceives the avatar as very happy, a rating of 6 means the avatar is perceived as very sad.

### 3.2.2. Survey to collect user ages and personally customized avatars

This survey was made in Google sheets. The survey will consist of three parts: an introduction, user age and avatar creation. Participants will be introduced to what this research is and what the true goal is: finding a correlation between user age and avatar characteristics. The introduction is also important for participants to understand a specific context given to them for which they create their personal avatar. The context that will be given is creating an avatar for a massively multiplayer online role-playing game (MMORPG) and interacting with friends you know from the real world. This context was chosen because the user is more likely to care about the way they represent themselves in that world (Hepperle, 2021; Lin & Wang, 2014; Paquette, 2013). Participants will also be asked to provide their age along with a screenshot of their personally created avatar from Ready Player Me (RPM). A link to the avatar creation site on Ready Player Me (RPM) will be provided on the survey. The resulting list of created avatars are found in appendix 15.

### 3.2.3. Survey to rate customized avatars

The survey to rate customized avatars was conducted after the previous survey was finished with an entirely different sample group of participants. Participants will be asked to rate avatars from the previous survey based on the superset created by the interviews. An example question is shown in appendix 8. The survey asks participants to rate one avatar, however, participants are allowed and told from the introduction that they are free to take the survey more than once. This is to garner more ratings from interested participants. Each avatar will need a minimum of two to three ratings on each of the characteristics to be able to assess consistency in the rating on that characteristic. Rating of avatars puts participant's construct through a narrow grid on how they, in general, differentiate between avatars (Walker & Winter, 2007; Wright & Cheung, 2007). The resulting ratings will quantify the avatar characteristics into numbers for data analysis to see if there is an established correlation between user age and avatar characteristics.

## 3.3. Material

To analyze the participant responses and perform mathematical assessments of the multi-variable correlation is done with Python, a popular programming language with appropriate plugins.

Ready Player Me is also necessary as part of the data collection. Ready Player Me (RPM) is an Estonian-based company that offers a free avatar customization site where users can create their own personal VR avatars that can be transposed onto other VR video games. These video games, of course, need to be compatible with the RPM game engine. Such game examples include VR chat, Nemesis, Wonda VR, Sidequest.

The avatars that participants from the interviews were exposed to was a list of created avatars that were generated by RPM. The list of avatars are numbered and listed in appendix 14. The participants that rated avatars were based on the avatars created in RPM by participants from the first survey. That list of avatars generated can be found at appendix 15.

### 3.4. Setting

All surveys were conducted remotely. The interviews were conducted in both a face-to-face or online setting.

### 3.5. Sample

To limit the amount of data needed, the research aim is to get enough data to run statistics (python) for a proof of concept rather than conclusive proof (Gonfalonieri, 2019). The standard sample for a conclusive correlation research would require a collected dataset of around 10,000 participants (Alkan & Daly, 2020; Brühlmann et al., 2020; Wei et al., 2017).

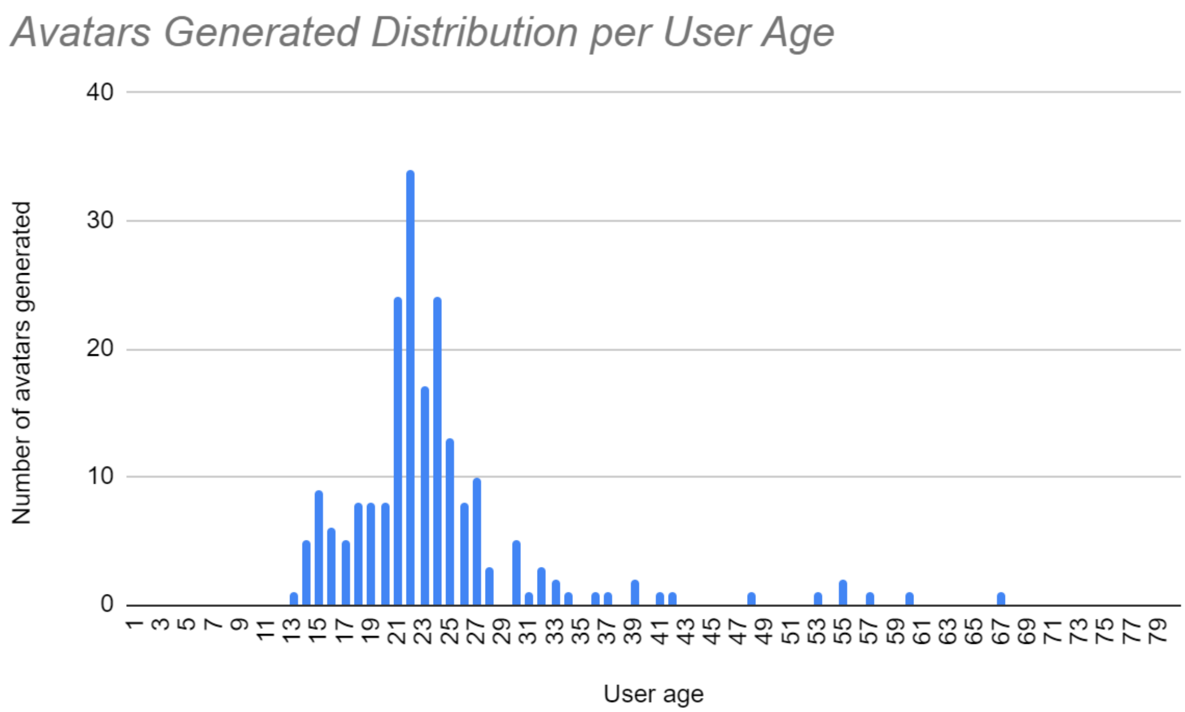
To determine the superset repertory, a sample of 10-15 responses provided a direction on the most important avatar characteristics (Fong & Mar, 2015; Gruijsters, 2020).

To collect a sufficiently large set of customized avatars in combination with the user and personally created avatar, a minimum of 100 participants is required. Users are meant to give their age and a screenshot of their personally created avatar from Ready Player Me (RPM).

To rate the created avatars on their avatar characteristics, another survey collected ratings from 2-3 different individuals on each of the characteristics from the superset repertory grid for each avatar. The data obtained in this survey will be conducted online via anonymous participants. This is needed to give an overall, averaged, quantification/impression of how an avatar is perceived per characteristic. Hence, the survey requires a total number of ratings that is two to three times the number of avatars times the number of characteristics generated from the previous survey. If the first survey generates 100 avatars, and if there are 20 characteristics per avatar, this survey needs to generate a total amount of 4000 - 6000 ratings (40-60 ratings per avatar).

The sampling for the interviews is completely randomized in order to make a repertory grid that identifies what people, in general, use to differentiate avatars. The target group for this current research are participants of all ages to better gauge an overall perception of avatars. The resulting age distribution of users that generated avatar is illustrated at Figure 2:

**Figure 2**




### 3.6. Instrument of data collection

Qualtrics and Google Sheets, these programs will facilitate all created surveys that will be sent out to participants. These programs are needed for statistics, organizing answers and data analysis.

### 3.7. Procedures for collecting the data

The procedure for collecting the data starts with the interviews, ergo, establishing the commonly referred to avatar characteristics. This is a set of online interviews where users are asked to describe the main differences they see between pictures of pre-created avatars. The resulting super-set repertory grid is made after transcribing all of the interviews and highlighting all the relevant themes to avatar characteristics.

The second phase of the data collection process is based on a survey with questions to capture each participant's age and their own personally created avatar from Ready Player Me (RPM). The survey will be designed where participants will create their avatar first, through a hyperlink on the survey to the Ready Player Me website, before giving their age and a screenshot of their personally created avatar. It will be required that pictures of their created avatar be uploaded in the survey.

The third phase has conducted a second survey to rate the created avatars from the previous survey. The participants will be given one avatar to rate through the lens of the superset repertory grid made from the interview. For example: “please rate avatar # on a scale of 1-6 between the following characteristics that best describes it? (happy vs. sad)”. A rating minimum of 2-3 per avatar is needed. These ratings will be saved and logged on Qualtrics and an Excel spreadsheet. Each individual avatar rating will be matched with the avatar creator’s age for better processing for the prediction model.

All the data from the both surveys will be organized into a dataset created on a CSV file that matches each individual rating to the avatar’s user age. This can be found at the link at appendix 2. The CSV file will be the key to answer to whether or not there is a correlation between user age and avatar characteristics. This will be done by finding the correlation matrix, testing the dataset through a linear regression model and exploring other options to conclusively establish a correlative relationship. Ultimately, this current research’s dataset is meant to be a number of user created avatars where the avatar characteristics are rated on the most frequently used characteristics, and the users age.

Where and how these surveys will be sent is by way of social media applications. These surveys will be shared on forums and platforms that are available such as Facebook, Instagram, Reddit, Steam, SurveySwap, WhatsApp and Discord.

### 3.8. Procedures for analyzing the data

The data analysis procedure is to determine correlation using linear regression analysis. The first interviews will determine the most commonly used characteristics to describe an avatar and create a repertory superset. The user generated avatars will be rated against the characteristics from the superset on a scale from 1 to 6 (1 meaning the avatar is seen to lean strongly to one side of the opposing spectrum representing a particular characteristic. 6 meaning the avatar is seen to lean strongly to the other side of the spectrum. 3 or 4 means that the avatar is not leaning strongly between either characteristic, suggesting that the characteristic is not strongly present.)

The dataset is then tested for correlation on a linear regression set. The x-variables used will be the rated avatar characteristics per avatar to find the y-variables (user age). This is a typical process for testing for accuracy in finding correlation (Brooks & Thompson, 2010; Waljee et al., 2019).

The statistical procedures to assess the level of correlation between user age and avatar characteristics is done with Python.

### 3.8.1. Finding correlation between user age and avatar characteristics

To determine whether it is possible to find a correlation between the user's age (y-variables) based on personally chosen avatar characteristics (x-variables), the use of linear regression is necessary as the first level of assessment (Chen, Kaufmann, 2021; ActiveState, 2022). If there is sufficient correlation between the quantified avatar characteristics and the user's age, a linear regression model will demonstrate the principle which then can be used for predictions by service providers (Ahmad, 2022; Mahlmann et al., 2010).

Since the approach is correlational research, the most important factor in this study is collecting user age and personally chosen avatar characteristics. The codes for python were found via online sources and experts in python (Malik, 2022; Frost, 2022; GeeksforGeeks, 2022a; GeeksforGeeks, n.d.; *Glossary of Common Terms and API Elements*. n.d.; *Pandas - Data Correlations*, n.d.; pandas.DataFrame.plot — Pandas 1.5.0 Documentation, n.d.; Python Machine Learning Multiple Regression, n.d.; Sifa, Drachen & Bauckhage, 2018; Stratogoly, 2022; sklearn.cross_decomposition.CCA, n.d.; *sklearn.linear_model.LinearRegression*. N.d.; *sklearn.model_selection.train_test_split*, n.d.; Yuen, 2020).

The validity of correlation will also be tested for the effects of the size of the data-set by randomly selecting subsets from the available avatars. This random selection of avatars is needed to find consistent average correlations (*coefficient of determination*). This has been shown in the random repeated tests.

#### 3.8.1.1. Why linear regression?

Linear regression was chosen as a first assessment of any predictability of user age through their avatar's characteristics. Linear regression is successfully used in many applications and there are many tools available in making it very accessible to a wide range of researchers

(About Linear Regression | IBM, n.d.; H20 AI, n.d.; Joby 2021). The quality of a linear regression fit is determined by the R-squared value. R-squared (*R^2, coefficient of determination*) is a statistical measure in a regression model that determines the proportion of variance in the dependent variable (predicted variable) that is found by the independent variables (Corporate Finance Institute, 2022; sklearn.linear_model.LinearRegression. n.d.). If R^2 equals 1, it is a perfect fit and a found correlation, meaning that there is a linear relationship between the x values and the y values, allowing to predict y (user age) given x (avatar characteristics) in all cases. If R^2 is zero, then there is no linear regression fit, meaning there is no linear relationship between the x and y-variables. Linear regression is a good first assessment before using non-linear models as non-linear relationships typically contain a linear component. Nonlinear models can typically be used as a further refinement of some level if predictability is based on linear models. If there is no linear correlation at all, it is unlikely that other nonlinear models will succeed as they too have a linear component (Microsoft, 2021; Sharma, 2022).

Please note, the assumptions underlying a linear regression model are as follows: (Joby, 2021; Kanade, 2022)

1. All the x-variables are independent and y-variables are dependent
2. There is no relation between the x-variables
3. The residuals (difference between the observed and predicted values of the dependent variables) have equal variance on every level
4. All residuals are have normal distribution

#### 3.8.1.2. Methods to finding linear regression model

Perfect or highly correlated relationships are found if the *coefficient of determination* is close to 1. (ActiveState, 2022; Dekha, 2022). The methods to find linear regression in this current research will be using the correlation matrix, principal component analysis (PCA) and cleaning of the data in order to best find a linear regression correlation (Boyle, 2005).

The correlation matrix is needed to gauge not only the correlation between the user age with the avatar characteristics but also the relationship between the avatar characteristics. This is to test whether all the x-variables are independent from each other. If there is a correlation between the x-variables, then PCA is required to make all the x-variables independent.

The purpose of PCA is to isolate the dependent x-variables and construct new variables which are not correlated in order to support the underlying assumptions for linear regression. PCA fits and compresses different components (avatar characteristics) through a standardization into fewer independent x-variables components (Galarnyk, 2017; Gruijter, 2020; Geeksforgeeks, 2022b; Sharma, 2022). The reduction of variables can also lead to loss of valuable information in the dataset. The variation ratio determines how much of the original data is retained by the number of components. In machine learning, retaining around 90% or more of the original data is a good rule of thumb for accurate testing (Sharma, 2020).

Even if there is a good linear relationship, there are two additional reasons why a particular dataset would not reveal this, even after the x-variables have been constructed to be independent with PCA: One is due to data-size (breadth of age groups, amount of ratings per avatar), the other is due to noise. Noise refers to corrupted/ inaccurate data from the dataset (Saseendran et al., 2019; Mishra, 2021). In the context of this current research, this happens if users lie about their true age, lie in their avatar ratings, or there are missing avatar characteristics needed to find a best fit. Linear regression assumes that the data has no noise and therefore any inaccurate rating will confuse the results. Cleaning the data is meant as sifting out any noise from the dataset (Neo, 2021). Therefore, cleaning the data is necessary to these inaccuracies to find better the linear regression result. The cleaning of the dataset will be done by outlining any variance difference between ratings per avatar of 3 or 10. This variance difference is meant to remove any individual outliers of ratings that is very much far removed from the rest of the ratings on the same avatar.

### 3.9. Ethical Considerations

All participants will be informed of the purpose and goal of this research study. Before the beginning of the interviews, participants will be asked to give consent that their answers be recorded and whether they wish to remain anonymous. The anonymous participants in the interviews are differentiated from the non-anonymous ones via the lack of a surname. Participants in all surveys will be allowed the choice to remain anonymous to ensure their own privacy. Any underaged participants need not to be treated differently as the instructions in the surveys are simple and understandable.

# 4. Findings

## 4.1. The data collection

The interviews had 12 participants recorded and interviewed. The avatar characteristics were extrapolated by how each participant differentiates between the avatars they have chosen. The characteristics are highlighted in the transcripts at appendix 16. The avatar characteristics found in the interviews are mentioned in Table 1 below.

**Table 1:**

*List of Avatar Characteristics found from the Interviews.*

| Categories: | Avatar characteristics mentioned in the interviews |
|---|---|
| Group of unique responses | Mechanical vs. human<br>Conformist vs. rebel<br>Hero vs. villain<br>Authentic vs. fake<br>Free vs. restrained<br>Idealistic vs. pessimistic<br>Special (superpower) vs. ordinary<br>Fantasy vs. reality<br>Attractive vs. unattractive<br>Futuristic dress vs. old-fashioned dress<br>Bright color clothes vs. dark color clothes<br>Productive vs. passive<br>Young vs middle-aged |
| Groups of common responses per avatar characteristic | Confident vs. shy<br>Know-it-all vs insecure (confidence)<br><br>Serious vs. relaxed<br>Professional vs. casual<br>Non-humorous vs. humorous<br><br>Neat and tidy vs. disorderly appearance |

| | |
|---|---|
| | Groomed vs. neglected<br>Tacky vs. Refined<br>Tasteful vs. tasteless |
| | Sticks out vs. blends in<br>Anonymous vs. identifiable |
| | Flamboyant appearance vs. restrained appearance<br>Theatrical vs. unpretentious appearance<br>Over-elaborate vs. understated appearance<br>Hip style vs. non-hip style |

- Note: The underlined avatar characteristics signifies that they were chosen for the superset repertory grid.
- Interdependent avatar characteristics were chosen from the same overlapping groups due to further gauge how a rater perceives the generated avatars used in this current research at appendix 15.

The resulting superset repertory grid made from Table 1 is as follows in Table 2:

**Table 2**

*Superset-Repertory Grid*

| Avatar characteristics: |
|---|
| Mechanical vs. human |
| Conformist vs. rebel |
| Hero vs. villain |
| Authentic vs. fake |
| Free vs. restrained |
| Idealistic vs. pessimistic |
| Special vs. ordinary |
| Fantastical vs. reality |
| Confident vs. shy |

Attractive vs. unattractive

Futuristic dress vs. old-fashioned dress

Bright color clothes vs. dark color clothes

Groomed vs. neglected

Tacky vs. Refined

Sticks out vs. blends in

Flamboyant appearance vs. restrained appearance

Theatrical vs. unpretentious appearance

Professional vs. casual

Productive vs. passive

Young vs. middle-aged

The 1st survey to collect newly designed avatars and the age of the creators, had 231 respondents. 212 pairs of avatars and user ages were usable to rate on the 2nd survey. The unusable responses were either corrupted, grainy or irrelevant to the research.

The 2nd survey to rate avatar characteristics through the superset repertory grid garnered 841 responses. Every avatar receives a minimum of 2-3 ratings from different participants on each of the characteristics found in the superset repertory grid. The questionnaires of both surveys are found at appendix 8 & 9. The data sets are transferred to Python as CSV files, which can be found at appendix 2.

### 4.2. Linear regression test

An initial linear regression test on the prediction model found no correlation between user age and personally chosen avatar characteristics from linear regression equation. Further testing is required. The codes used are found in appendix 3.

### 4.3. Correlation matrix

The correlation matrix was made to find the relation between all the avatar characteristics and user age. The full correlation matrix is found at appendix 10.

**Table 3**

*Avatar Characteristics' Correlation to User Age*

| Avatar characteristics | Correlation value to user age |
|---|---|
| Mechanical vs. human | -0.06 |
| Conformist vs. rebel | 0.00 |
| Hero vs. villain | 0.12 |
| Authentic vs. fake | 0.03 |
| Free vs. restrained | 0.04 |
| Idealistic vs. pessimistic | 0.10 |
| Special vs. ordinary | 0.08 |
| Fantastical vs. reality | 0.08 |
| Confident vs. shy | -0.03 |
| Attractive vs. unattractive | 0.11 |
| Futuristic vs. old-fashioned | -0.06 |
| Bright-color clothes vs. dark-color clothes | 0.01 |
| Groomed vs. neglected | 0.07 |
| Tacky vs. refined | 0.01 |
| Sticks out vs. blends in | 0.06 |
| Flamboyant vs. restrained | 0.00 |
| Theatrical vs. unpretentious | 0.03 |
| Professional vs. casual | -0.06 |
| Productive vs. passive | 0.02 |
| Young vs. middle-aged | 0.21 |
| User age | 1.00 |

This column is a correlation value to user age, there is a very low correlation between user age and each of the avatar characteristics. The overall correlation matrix in appendix 10 shows some dependency between the x-variables. This means that PCA is needed. Other methods such as *cleaning the raw dataset* and the selection of *most significant variable*s are used to reduce random noise which could affect the ability to find a strong correlation. Cleaning the dataset removes inconsistent ratings. The selection of most significant variables refers to selecting those x-variables (avatar characteristics) showing the highest correlation with the y-variables (user's age).

### 4.4. PCA

PCA will make all the avatar characteristics (x-variables) independent from each other. The PCA correlation matrix can be found at appendix 11. Comparing the PCA matrix and correlation matrix indicates the x-variables, that are based on the rated avatar characteristics, are independent and now can be used in the linear model.

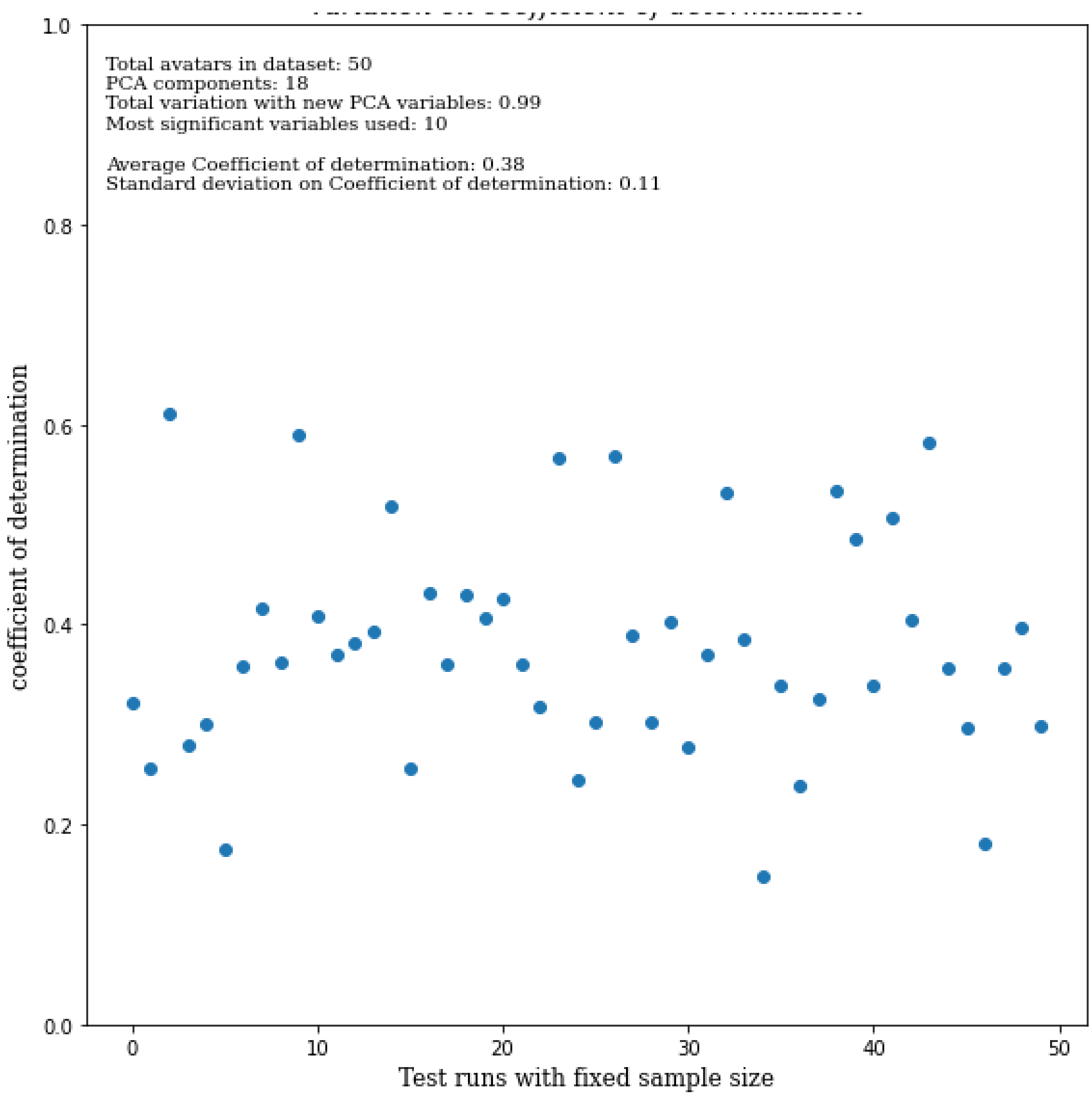
### 4.5. Determining the influence of 'noise' in the data

The *cleaning of this current research's dataset* is needed to extrapolate whether or not it is noise affecting the value reduction in the *coefficient of determination*. This is done by filtering out avatars that demonstrate a large variation on the ratings per avatar characteristic. As a first approach, on a scale rating of 1-6, if an avatar has a variation of more than or equal to 3 points on 10 or more of its characteristics, it will be removed from the dataset. This selection has reduced the number of avatars from 208 to 197 avatars. The effect of noise will be tested through different sample variations in sample size with via 50 and 70 avatars over 50 tests.

By repeatedly taking a repeated random selection of 50 avatars out of 197 avatars, It was found that there is a very large variation in *R*^2 in Figure 3: *average*: 0.38 with a *standard deviation* of 0.11.

**Figure 3**

*Variation on Coefficient of Determination*



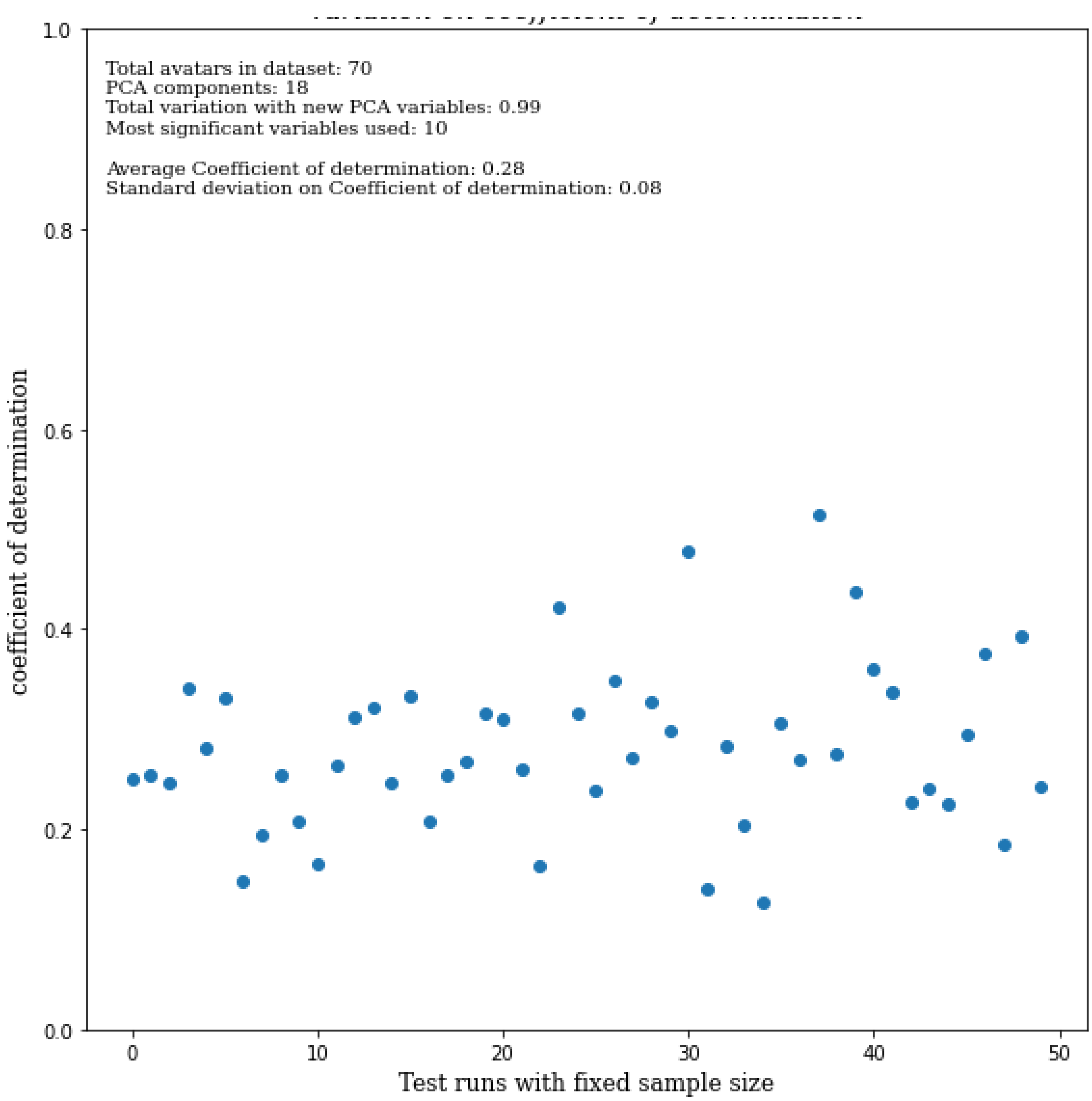

- Note: this figure is demonstrating the average *correlation (coefficient of determination)* from 50 tests of random repeated selection of 50 avatars out of 197 avatars. Each test is a dot on the graph.
- The variables are described as follows:
    - **Total avatars in dataset:** number of randomly chosen avatars used per test (dot)
    - **PCA components:** number of x-variables changed to independent
    - **Total variation with new PCA variables:** percentage of how much of the information from the original x-variables is retained with the new PCA variables

- **Most significant variables used:** number of PCA variables used in the test who have the highest correlation to user age.
- **Average coefficient of determination:** average correlation value found through 50 selections (test runs) of 50 randomly selected avatars
- **Standard deviation on coefficient of determination:** the variance across the coefficients found with each test run

By taking a repeated random selection of 70 avatars out of 197 avatars in Figure 4, there is a smaller variation of *R^2* with a lower average: *Average:* 0.28, *Standard deviation:* 0.08.

**Figure 4**

*Variation on Coefficient of Determination*

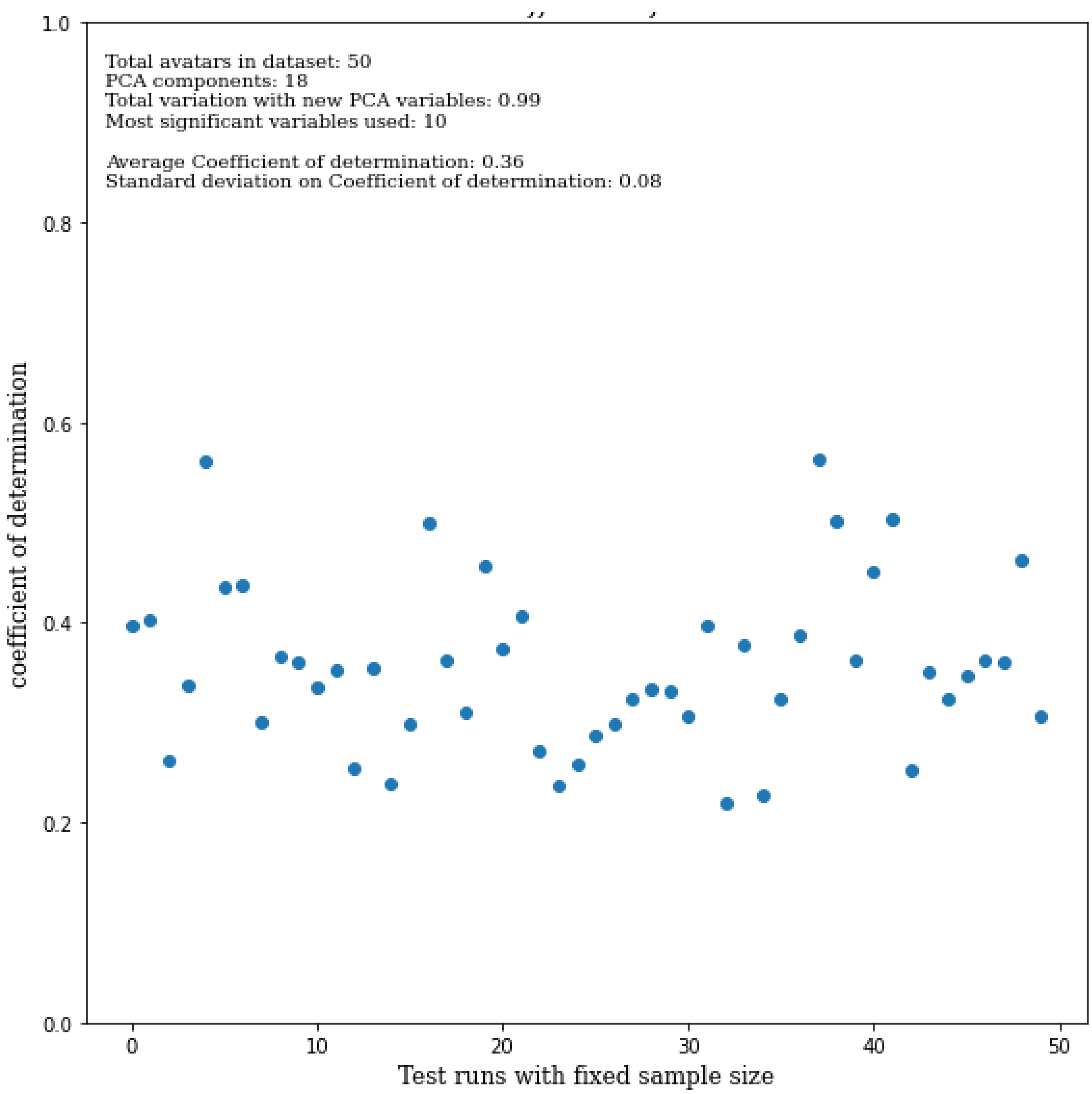


- Similar variable description as **Figure 3**
- **Average coefficient of determination:** average correlation value found through 50 selections (test runs) of 70 randomly selected avatars out 197 total avatars.

Another attempt was made by removing avatars who showed a variation of more or equal to 3 points on 3 or more avatar characteristics. This selection of avatars has reduced the number of avatars from 208 to 82 avatars. By testing out a repeated random selection of 50 avatars from this 82, the average *R^2* of 0.36 and standard deviation 0.08. This is illustrated in Figure 5.

**Figure 5**

*Variation on Coefficient of Determination*

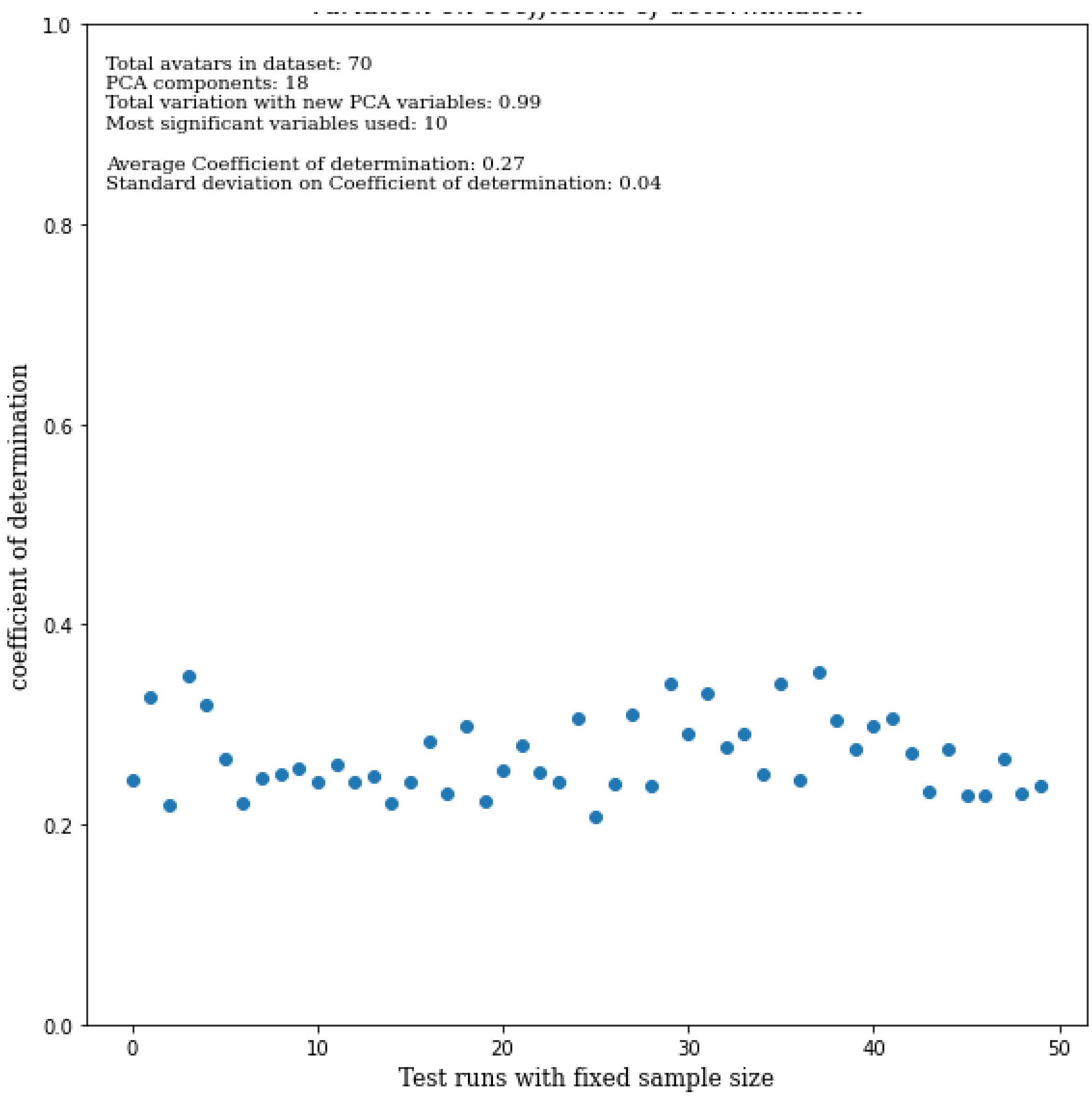


- Similar variable description as **Figure 3**.
- **Average coefficient of determination:** average correlation value found through 50 selections (test runs) of 50 randomly selected avatars out 82 total avatars.

The next test was a repeated random selection of 70 avatars out of 82 avatars in Figure 6. The average *R*^2 was 0.27 with a standard deviation of 0.04.

**Figure 6**

*Variation on Coefficient of Determination*



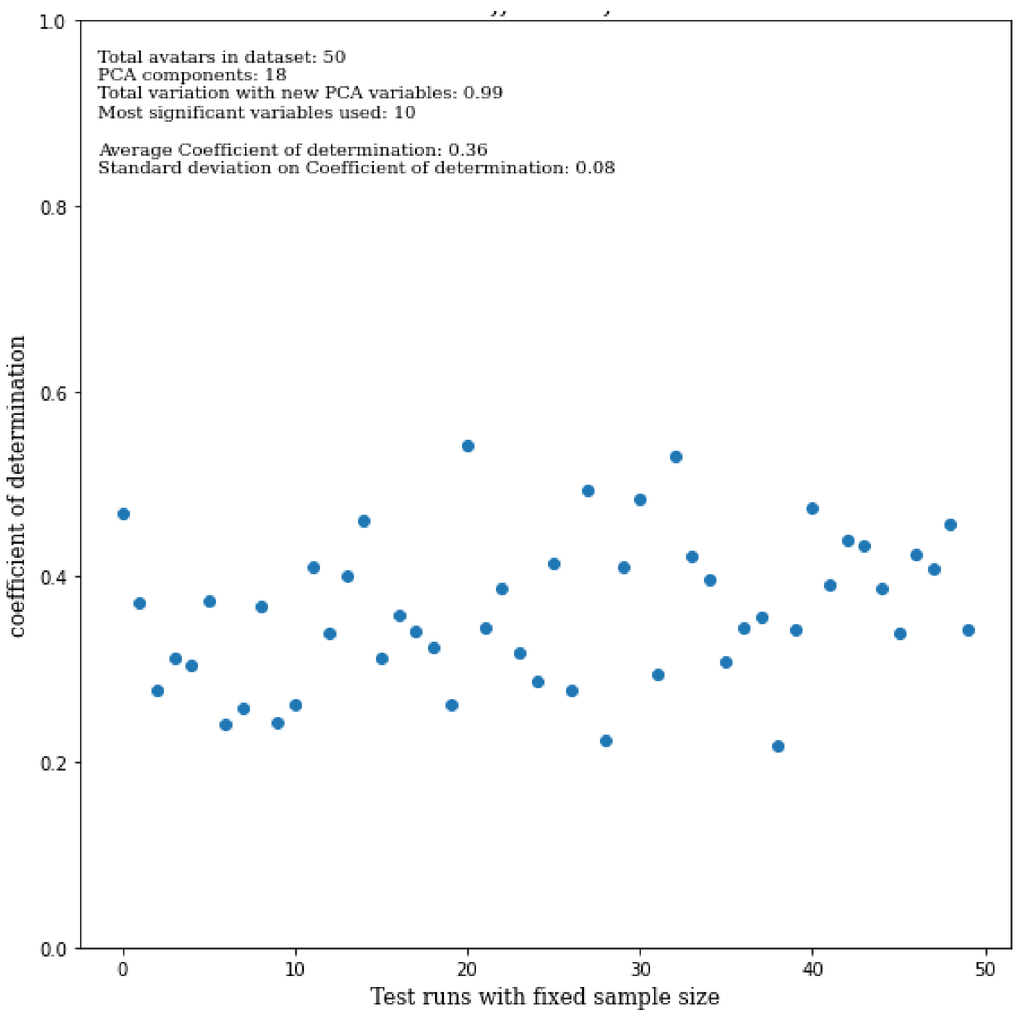

- Similar variable description as Figure 3.

Looking at the data above, it is clear that the average value of the coefficient of determination is not affected by a more stringent selection of the dataset with a given random size (sample size of 50: 0.38 vs. 0.36) (sample size of 70: 0.28 vs. 0.27). The variation of the *coefficient of determination* is larger when we apply a less restricted selection of the dataset.

Taking a sample of 50 or 70 avatars from a larger dataset brings more variation in the selected samples which leads to a larger variation of *coefficient of determination*. To concept is illustrated below in Figure 7 with a less restricted data selection from 197 to be forced under 82 avatars with a 50 repeated random samples:

**Figure 7**

*Variation on Coefficient of Determination*




- Similar variable description as Figure 3.

The standard deviation on the *coefficient of determination* is 0.08, as was seen in Figure 5. The *coefficient of determination* is high, but that is due to the effect of a randomly selected 82 avatars out of 197 that I used to test the effects of standard deviation. Apparently the 82 avatars selected from the 197 accidentally showed a fairly good *coefficient of determination*, but nothing outside the range that was seen in earlier samples drawn from the 197 avatars (see Figure 4). The minor differences between the standard deviation on the *coefficient of determination* shows that the noise is not the contributing factor to the differences seen before. Instead the size of the dataset drove the difference in the standard deviation. The average coefficients show that the noise is not the problem but the size of the resulting dataset.

Based on the findings from above, the cleaning method to create a more restricted dataset did not fundamentally shift the average *coefficient of determination* with a given random sample size (sample size of 50: 0.38 vs. 0.36, with a sample size of 70: 0.28 vs. 0.27). A larger sample-size reduces the average *coefficient of determination* in (0.36 to 0.27). This is due to the effect where a smaller number of observations tend to result in a higher *coefficient of determination* in the linear regression. Based on this, it is likely to assume that a larger dataset will further reduce the *coefficient of determination*, thereby suggesting that there is no correlation between user age and personally chosen avatar characteristics.

A more restricted, clear dataset does not affect the standard deviation of *coefficient of determination* from each repeated random size.

### 4.6. Determining the influence of sample size

The next step is to determine whether the sample size has a significant effect on the coefficient of determination. When looking at *R^2* (*coefficient of determination*), there is a noticeable reduction in value based on the increasing volume of avatars selected. This could be due to the fact that less data-points make it easier to draw an optimized line. For instance, with only two data-points, there is going to be a perfectly fitting line, with more data points, it is going to get harder to find a perfect fit line.

In the previous paragraph it was determined that the noise in the data does not have a significant effect on the coefficient of determination. To determine effects of the sample size, a looser data selection can be used to see how the coefficient of determination changes with the sample size. In Table 4, the data selection is restricted based on removing avatars that have more than or equal to 3 point differentiation in 10 or more avatar characteristics. This reduces the dataset from 208 to 197 avatars. The repeated random selection to demine the average coefficient of determination as a function of the sample size is as follows:

**Table 4:**

*Change in Coefficient of Determination as a Result of Sample Size*

| Sample size | Coefficient of Determination |
|---|---|
| 50 | 0.37 |
| 80 | 0.25 |
| 100 | 0.21 |
| 140 | 0.17 |
| 190 | 0.14 |

An increasing sample size shows a lower correlation value (*coefficient of determination*) between user age and avatar characteristics. It can therefore be concluded that the larger the sample size of avatars, the less correlation is found. Hence making the hypothesis of this current research to be not confirmed.

### 4.7. Additional observations

The rated avatar characteristics are meant to extrapolate what the user wants to present to themselves as. The results below are a summary of the rating scoring, on average, on how all age groups were rated as. The full details of these average scores can be found at appendix 12. These additional observations are meant to sift out specific trends and meanings in how overall age groups compare with each other.

**Table 5**

*Overall Average Scoring of Avatar Characteristics in all Age Groups:*

| Categories: | Avatar characteristics: |
|---|---|
| Equal distribution across both characteristics | Professional vs. casual, sticks out vs. blends in |
| Near equal distribution across both characteristics | Conformist vs. rebel, special vs. ordinary, fantastical vs. reality, futuristic vs. old-fashioned, bright-color clothes vs. dark-color clothes, theatrical vs. unpretentious, flamboyant vs. restrained |
| Distribution concentrated in the middle (3-4) | Productive vs. passive, hero vs. villain, authentic vs. fake, free vs. restrained |
| Preference to a specific avatar characteristic from all age groups | Young vs. middle-aged, free vs. restrained, hero vs. villain, idealistic vs. pessimistic, confident vs. shy, groomed vs. neglected, mechanical vs. human, tacky vs. refined |
| Major trend deviation between age groups | Attractive vs. unattractive, Fantastical vs. reality |

- This table is a summary of the most notable trends from the graphs in appendix 12

Table 5 shows that the majority of users prefer to present themselves as more under the characteristics like human, free, hero, idealistic, attractive, confident, groomed and refined.

It can be observed in the attractive vs. unattractive that the average scoring rating went progressively to more unattractive scores the older the user was. Fantastical vs. reality shows that the older the user is, the more they prefer to present themselves as more realistic compared to the varied younger age groups.

Since a majority of the users that generated avatars were people aged between 10-38, Gen Z and millennials, there is more insight in how they wish to be perceived than any other age group. Table 6 below is a list of the most notable features per characteristic. Table 6 is based on the graphs that can be found in appendix 13.

**Table 6**

*Average Age Groups (10-38) Rating per Avatar Characteristic:*

| Categories | Avatar characteristics | Relevant age range to trend in avatar characteristic |
|---|---|---|
| Age groups with equal distribution across both characteristics | Conformist vs. rebel | 10-29 |
| | special vs. ordinary | 10-29 |
| | fantastical vs. reality | 20-26 |
| | professional vs. casual | 10-26 |
| | bright-color clothes vs. dark-color clothes | 10-29 |
| | sticks out vs. blends in | 20-29 |
| Age groups with near equal distribution across both characteristics | Authentic vs. fake | 17-29 |
| | futuristic vs. old-fashioned | 10-29 |
| | tacky vs. refined | 20-29 |
| | productive vs. passive | 10-23 |
| | flamboyant vs. restrained | 10-29 |
| | theatrical vs. unpretentious | 17-26 |
| Age groups with specific avatar characteristic preference | Mechanical vs. human | 10-29 |
| | hero vs. villain | 10-26 |
| | free vs. restrained | 17-29 |
| | confident vs. shy | 20-26 |
| | idealistic vs. pessimistic | 20-26 |
| | attractive vs. unattractive | 32-38 |
| | groomed vs. neglected | 17-29 |
| | young vs. middle-aged | 10-35 |

- Note: this table is a summary of the most significant trends found between the age groups (10-38) as well as specifying the range between ages that the age trend is.

The majority of this sample prefer to present themselves as human, heroic, free, confident, idealistic, attractive, groomed and young. The majority of trends stem from users in their 20s who have the biggest variation in preference across both opposing characteristics. The overall change in preference per avatar characteristic was mostly found in users aged older than either 26 or 29 years old (Brunjes, 2022). Analysis on the young vs. middle-aged reaffirms the same trend as Table 5: the older the user, the less attractive they are rated. Users that are younger than 30 have more variation in avatar characteristic preference than older users. But this needs to be confirmed with a larger, more equally distributed dataset. The python codes used in the findings can be found at Appendix 3-7

# 5. Discussion

The hypothesis to be tested was that there was a strong correlated relationship between user age and personally chosen avatar characteristics. This hypothesis was not confirmed in the findings. It has been found that a larger dataset will further reduce any coefficient of determination, which suggests that there is no real correlation to be found between the user age and their personally chosen avatar characteristics. It is not due to the noise of the data as, after cleaning the dataset, it has not fundamentally shifted the average coefficient of determination. The highest correlated avatar characteristic to user age from the correlation matrix was “young vs. middle-aged” as 0.209445. The lowest correlated characteristic to user age was “futuristic vs. old-fashioned” as -0.0602473. These characteristics as well as the others have indicated a weak correlated relation to user age. More details can be found at appendix 10. This current research’s focus on studying the effect of age on the way people design their avatars is not dominant. In other words, people’s own constructs leading to their own avatar design choices is not strongly determined by their age (Carver & Scheier, 2000; Dechant et al., 2021; Oosterwegel, 1995). Since there was no linear correlation found, the prediction modeling potential to discern a user’s age from their chosen avatar characteristics is no longer possible. It is very possible that a user’s personalized avatars can be used to predict other user-related variables. Examples could include a user’s behavior, personality type and possible exposure to video games (Hadiji et al., 2014; Lawton, Carew & Burn, 2022). Past research has found connections between avatar customization motives and individual’s personalities on scales like extraversion, agreeableness and neuroticism (Fong & Mar, 2015).

While this research did not find a correlation between the avatar characteristics and the age of the user creating the avatar, there are a number of valuable aspects resulting from this research which can be used by service providers offering online gaming platforms. To start with, avatars are the key to initial engagement to a platform for new users (Nadeem et al., 2020; Trepte & Reinecke, 2010): If users are allowed to have a large variety of the options to personalize avatars to their own sense of identity, the more loyal they'll be to the platform (Ducheneaut et al., 2009; Fong & Mar, 2015; Koles & Nagy, 2012; Kristanto, 2018; Nadeem et al., 2020; Wu, 2014; Yee et al., 2009). Strong loyalty and large time spent on the platform allows service providers to better profile their users. The tracking of actions on customized avatars on these platforms could create a user's profile. It also opens the possibility to monetize off a users' desire to personalize their avatars through the sale of new avatar characteristics (Böffel et al,. 2022). The sale of avatar characteristics can be extended via the building of a community of like-minded users. Personal construct psychology has shown that positive relations are built on similarities in beliefs and interests. Hence, such a community would validate a user's self-created avatar identity which correlates with an increased avatar customization and engagement to the platform. (Ducheneaut et al., 2009; Mawhorter et al., 2018; Wu, 2014).

Another issue for service providers of a successful gaming platform is to ensure a good user experience. A good user experience helps determine how much a user will be engaged on the platform (Ducheneaut et al., 2009; Fox & Ahn, 2012; Koles & Nagy. 2012). Therefore knowing what ensures a good user's experience is crucial. A repertory grid is a good multi-purpose assessment tool to create a general mindmap that people use to differentiate between objects such as avatars, or their user experience in using a gaming platform. (Oosterwegel, 1995; Walker & Winter, 2007). Analytics and service providers can use repertory grids to do competitive analysis between different gaming platforms to see what provides the best user experience (Alkan & Daly, 2020; Kelly, 2017; Klöckner & Prugsamatz, 2012). If this research is repeated over time, changes in user expectations and preferences could be found, which offers the opportunity to design games that anticipate future appeal.

The additional observations have found age group trends per avatar characteristics from Table 5 and 6 and the graphs at appendix 12 and appendix 13. This is limited in participant intake as most of the participants were aged between 10 and 38 years old. However, this may provide the groundwork towards insight on how users like to present themselves as avatars per age demographic to service providers. Since there are more participants between the ages of 10 and 38, the graph trends in avatar characteristic preference is more valuable (appendix 13) than graphs noting all the age groups (appendix 12).

There was a found division in preference between all age groups which began mostly with users getting older than their 20s. The age groups in the 20s had the biggest range in variation between the opposing avatar characteristics than any other age groups. This could imply that users in their 20's tend to use avatars as a means for  experimentation of identity in self-presentation to others (Hemp, 2014; Koles & Nagy, 2012; Lin & Wang, 2014). In the context of different generations, millennials in their late 20s have more similar preferences of presentation to Gen Z in their early 20s than older millennials in their 30s. Service providers can use this observation as confirmation that an avatar customization with a large variety of choices is useful for engaging more users onto their platform (Ducheneaut et al., 2009; Fong & Mar, 2015; Koles & Nagy, 2012; Yee et al., 2009; Wu, 2014). This is more apparent in regards to users in their 20s.

Based on the context given to participants in the first survey (MMORPG), all the age groups that generated avatars from tables 5 & 6 largely preferred to present themselves with avatar characteristics like human, free, heroic, idealistic, attractive, confident, groomed and refined (appendix 12). Service providers should ensure that they have these types of avatar characteristics available in their avatar customization tools.

It was also noted that most users presented themselves as in the middle of the opposing characteristics, either (3) or (4). These avatar characteristics were productive vs. passive, hero vs. villain, authentic vs. fake, free vs. restrained. This means that most of these participants, under the context of an MMORPG, don't have a strong preference to use any of these characteristics as a strong differentiation of their avatar.

On Figure 27 of "young vs. middle-aged", it was found that as the users got older, the more "middle-aged" the users were rated on average. There were naturally exceptions but this also echoes a sentiment that perceptions on what is considered "young" and "middle-aged" does change between generations. However, the number of participants older than 40 is not as many as the younger participants. The lack of equal distribution among all age groups puts any conclusion on age preference trends to be subjected as unreliable.

## Limitations:

In this study, there were various limitations found. The first limitation was the limited dataset with no equal distribution between the age groups as indicated in Figure 2. There is also a lack of avatars generated in every single age group. This current research received age groups between 10-70. This limited dataset also affects the conclusions that were drawn from the trends from the graphs and tables 5 & 6 and appendix 12. The current results make any insight found to be subject to scrutiny due to a lack of equal age distribution among all age groups.

Another limitation is the lack of verification on the age of users that generated avatars in this current research. This current research took a random sample from anonymous strangers on Reddit, Surveyswap, WhatsApp, Discord, Teams. It is similar to the rating of said avatars. However, the ratings were better verified through its attention checks and python methods such as PCA and the cleaning of the dataset. However, some screenshots of generated avatars were affected by poor quality image which affects how raters would've rated said avatar.

Since all of this data had to be made and logged in manually, there may have been some error in the ratings of avatars. Some ratings could've been pushed to a different avatar.

The context given to users to generate avatars was quite broad. The lack of a more specific game platform context is an important factor, as platform context informs the expectations and choices of users in customizing their avatars. One final limitation is the fact that RPM has limited avatar characteristics to customize such as no age factor to customize, this affects how users will create and rate avatars.

One other limitation relates to a lack of quantification of the level of the participant's exposure to video games in general. Based on the theory of Personal Construct Psychology, exposure and experience to video games affects the perception and choices in how they create their avatar. The level of exposure to video games was expected to be age related, but it was not explicitly asked from the participants who created avatars. This casts a blind spot on the conclusions of this current research on the extent of difference in customization between experienced and in-experienced participants. Perhaps, age was not as strongly tied to the level of experience a user had with video games, resulting in a weak correlation.

Other possible limitations could lie with a lack of accessibility due to the surveys having been written in English, which limits possible participants from those that cannot speak English.

Another limitation is the participants' honesty in their answers. A lack of honesty will muddle the accuracy of the prediction model. And since this is a survey conducted remotely to anonymous people, there is no proper way to verify the truthfulness of a participant's response.

# 6. Recommendations

The first recommendation is to acquire a bigger dataset sample with a more equal distribution across all age demographics to conclusively prove that there is no linear correlation. It is also recommended to increase the ratings collected per avatar. 2-3 ratings per avatar is light for any statistics, 5-6 ratings per avatar could shed a better light on the perceived values (Tenny & Abdelgawad, 2021). The repertory grid is a good basis for capturing a group of people's emotions/perspectives with independent avatar characteristics. A deeper dive in a larger, equal distribution in all age groups with more ratings per avatar could completely change the outcome.

This new large data sample could also be used to in/validate the data trends of age groups from appendix 12 & 13 since this current research has found no established correlation. This could potentially provide more insight on how to target a specific age demographic. Furthermore, I recommend trying out other types of modeling such as nearby neighbors, non-linear regression, decision trees (Gao, Lui & Wu, 2010; Ghorakavi, 2021; Koehrsen, 2017). Linear regression is a good indicator on whether there is a good fit for these models but that is never a complete certainty.

Since RPM offers only a limited set of options to personalize avatars, further studies should continue studying user perspectives on avatars with more varied avatar customization options. The intent is to find out on how characteristics such as aging differ from perspectives/constructs between age groups/ generations (Daignault, Wassef & Nguyen, 2021).

Another recommendation is to add more interdependent avatar characteristics to better gauge the similarities in emotional drivers between user's perspectives on avatars. This is due to the fact that there were found correlations between avatar characteristics in the correlation matrix: special vs. ordinary & sticks out vs blends in (0.59), flamboyant vs. restrained & sticks out vs blends in (0.61), flamboyant vs. restrained & theatrical vs. unpretentious (0.57). These avatar characteristics carry interdependent meanings, which indicate promise in finding a quantifiable way to realizing how people, in general, perceive objects such as avatars (Table 3, appendix 10). Further recommended research is to look into using a more dynamic avatar customization with more options to customize as well as how specific game platform context changes the choices of avatar for the user. It could even be tested on different mediums (mobile, computer, VR).

There is no age function to customize avatars on any of Ready Player Me's avatar customization menu. With the current tool, each avatar looks to have the same age. This leaves the current research to deduct a correlation with the user's age based only on the customization choices on clothes, hairstyle which is influenced by how each age demographic perceives these options. Past research has found that everyone has a different perception on what is considered "old" and "young" (Daignault, Wassef & Nguyen, 2021). Further research should be on further delving into this topic to see if there is any credence on a quantitative level. This proposed research could impact the perceptions of age and to what extent it is perceived as artificial.

Since there was a limitation in RPM avatars in avatar characteristics, specifically modifying the avatar's age, further studies should research the effects of this avatar age with avatar customization engines that have more options. Past research have found that the age of the user has a different perspective which influences their perception on the age of the avatar (Ducheneaut et al., 2009)

# 7. Conclusions

The insight obtained from researching for a correlation between a user's age and personally chosen avatars characteristics indicates that there is no correlation relationship from linear regression model. This proof of concept research is best recommended to other service providers and researchers to look into a larger collected sample to make a more conclusive conclusion on this research. The superset repertory grid is believed to be a good foundation to quantify characteristics of avatars as the noise in the data remained low.

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